

Q1 (multichoice)

Purpose of loops

What is the main purpose of using a loop in programming?

- ☐ To execute a set of statements only once
- ☐ To repeat a set of statements multiple times
- ☐ To compare two numbers
- ☐ To define a variable

Q2 (multichoice)

Loop types in Java

In Java, which of these is **not** a type of loop?

- ☐ while
- ☐ for
- ☐ do while
- ☐ repeat until

Q3 (multichoice)

What controls loop continuation

What is the control structure that allows a loop to continue or stop?

- ☐ Boolean condition
- ☐ String comparison
- ☐ Arithmetic operator
- ☐ Input statement

Q4 (multichoice)

While syntax

Which of the following is the correct general form of a **while** loop?

- ☐ while (condition) { statements }
- ☐ loop (condition) { statements }
- ☐ do { condition } while { statements }
- ☐ repeat (condition) { statements }

Q5 (multichoice)

LCV responsibilities

In a `while` loop, the loop control variable (LCV) must be:

- ☐ Declared but never updated
- ☐ Updated only after the loop
- ☐ Declared, tested, and updated inside or before the loop
- ☐ Automatically managed by Java

Q6 (multichoice)

Always-true while

What happens if the Boolean condition in a `while` loop is always `true`?

- ☐ The loop runs once
- ☐ The program crashes
- ☐ The loop runs infinitely
- ☐ The compiler rejects it

Q7 (multichoice)

Hello World while count

How many times will this code print "Hello World"?

```
int i = 1;
while (i <= 4) {
    System.out.println("Hello World");
    i++;
}
```

- ☐ 3
- ☐ 4
- ☐ 5
- ☐ Infinite

Q8 (multichoice)

For header visibility

Which parts are **explicitly visible** in a `for` loop header?

- ☐ Initialisation, condition, post-body action
- ☐ Only the condition
- ☐ Only initialisation and condition
- ☐ Only post-body action

Q9 (multichoice)

Infinite for

Which of the following `for` loop headers would create an infinite loop?

- ☐ `for ( ; ; )`
- ☐ `for (int i = 0; i < 5; i++)`
- ☐ `for (int i = 10; i > 0; i--)`
- ☐ `for (int i = 1; i <= 10; i++)`

Q10 (multichoice)

For update location

In a `for` loop, where is the **update of the LCV** typically located?

- ☐ Before the loop
- ☐ Inside the loop body
- ☐ In the third part of the loop header
- ☐ After the loop ends

Q11 (multichoice)

For output sequence

What output will this code produce?

```
for (int i = 1; i <= 3; i++) {
    System.out.println("Num: " + i);
}
```

- ☐ Num: 0, Num: 1, Num: 2
- ☐ Num: 1, Num: 2, Num: 3
- ☐ Num: 1, Num: 2, Num: 3, Num: 4
- ☐ None

Q12 (multichoice)

While vs Do-while difference

What is the main difference between a `while` loop and a `do while` loop?

- ☐ `while` runs once, `do while` runs many times
- ☐ `do while` runs its body before testing the condition
- ☐ `do while` can't use Boolean conditions
- ☐ There is no difference

Q13 (multichoice)

Do-while syntax

Which of the following is the correct syntax for a `do while` loop?

- ☐ `do { statements } while (condition);`
- ☐ `while { statements } do (condition);`
- ☐ `loop (condition) { statements }`
- ☐ `do while { statements }`

Q14 (multichoice)

When to choose do-while

When should you choose a `do while` loop instead of a `while` loop?

- ☐ When the body must run at least once
- ☐ When the condition should never be checked
- ☐ When the loop must never terminate
- ☐ When you don't need a counter

Q15 (multichoice)

LCV definition

What do we call a variable that controls how many times a loop executes?

- ☐ Sentinel variable
- ☐ Loop control variable
- ☐ Incrementer
- ☐ Boolean variable